Alex Bailo

Deep Learning · Computer Vision · 3D Reconstruction

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Summary

Technical skills: Python, C++, Pytorch, ONNX, OpenCV, Open3D, Git
Languages: Fluent in English, Ukrainian, Russian; advanced level in Korean; learning Dutch

Experience

Qualcomm — Staff ML R&D Engineer, XR team — *Amsterdam, Netherlands*
Jul. 2021 - present
- 3D reconstruction and scene understanding for AndroidXR
- Line manager

Oosto — Deep Learning Researcher and Scrum Master — *Belfast, UK*
Apr. 2020 - Jun. 2021
- Body detection and re-id
- Replay face spoofing attack detection

Kakao Brain — Deep Learning Research Engineer — *Seongnam, S.Korea*
Nov. 2019 - Mar. 2020
- Human pose estimation and action similarity research for fitness tracking

Noul Inc. — Computer Vision & Deep Learning Research Engineer — *Yongin, S.Korea*
Aug. 2017 - Oct. 2019
- Microscopy diagnosis of malaria. Development from training to edge product integration
- Created hematology analyzer with detection, segmentation, and classification capabilities
- Research on GANs for medical data augmentation resulting in publication

Healthrian — Software engineer, Intern — *Daejeon, S.Korea*
Jun. 2015 - Aug. 2015
- Developed an Android application for ECG medical device

My Design Lab • KAIST — Undergraduate Researcher — *Daejeon, S.Korea*
Dec. 2014 - Jun. 2015
- Developed a drone to implement wall painting works for skyscrapers

Education

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, S.Korea*
M.S. in Electrical Engineering. Robotics and Computer Vision [lab.] supervised by [In So Kweon] — *Sep. 2015 - Aug. 2017*
- A real-time vehicular vision system to seamlessly see-through cars
- Machine learning-based autonomous vehicle vision system

Korea Advanced Institute of Science and Technology (KAIST) — *Daejeon, S.Korea*
B.S. in Electrical Engineering & Business and Technology Management — *Sep. 2011 - Aug. 2015*
- Manager at KAIST International Basketball Club (KIBC)
- Vice President, Public Relations Head at KAIST International Student Association (KISA)
- Exchange in HKUST and Nanyang Technological University

Selected Publications

International Conferences

CVPR26 Fast SceneScript: Accurate and Efficient Structured Language Model via Multi-Token Prediction
ICCV23 DG-Recon: Depth-Guided Neural 3D Scene Reconstruction
CVPRW19 Red blood cell image generation for data augmentation using cGAN
ICCV17 VPGNet: Vanishing Point Guided Network for lane and road marking detection and recognition
WACV17 Robust road marking detection and recognition using density-based grouping and ML techniques

International Journals

CVIU22 MC-Calib: A generic and robust calibration toolbox for multi-camera systems
IJCV22 Real-Time Multi-Car Localization and See-Through System
Access21 A Body Part Embedding Model With Datasets for Measuring 2D Human Motion Similarity
PRL18 Efficient ANMS for homogeneous spatial keypoint distribution